

Background

Due to the lack of claims & severity data as well as portfolio information, the goal of this rater is to approximate premiums from first principles through severity and frequency estimation, and applying an expense & profit margin afterwards to obtain a premium.

Throughout the modelling, it was assumed that the only available information were the inputs stated in the case statement: USDOT, # drivers, power units (and VINS), operating radius, and hauled commodities.

Datasets & Sources

A primary source of data was through the Federal Motor Carrier Safety Administration (FMCSA) open data program. From the FMCSA, the datasets “Company Census File” and “Crash File” were used.

The company census file provided information on carriers by USDOT, such as mileage, # power units, # drivers, and state. This was filtered in the query to only include “for-hire” carriers. The crash file provided information on individual recorded truck crashes. These datasets were joined together to form a view on carriers and their prior crash experience.

For severity estimates, values were taken from the Texas Automobile Insurance Plan Association (TAIPA). Information surrounding commercial auto claim severity appeared scarce, thus this was one of the primary sources of obtaining an actual insurance payout value.

“Crash Costs for Highway Safety Analysis” from the Federal Highway Administration (FHWA) was used as a proxy for different severity levels of claims. This was combined with the information from TAIPA to get a proxy of claim severities for different crash severities.

To approximate crash severity distributions, data was used from the National Highway Traffic Safety Administration (NHTSA).

Information surrounding the different classes of trucks such as weight and possible models was taken from the Department of Energy of the US.

Estimation of expense ratios were taken from articles published by the Casualty Actuary Society (CAS).

Links

<https://data.transportation.gov/Trucking-and-Motorcoaches/Company-Census-File/az4n-8mr2>
<https://data.transportation.gov/Trucking-and-Motorcoaches/Crash-File/aayw-vxb3>

<https://www.tdi.texas.gov/rules/2026/documents/2026comautomemo.pdf>
<https://highways.fhwa.dot.gov/sites/fhwa.dot.gov/files/2022-09/fhwasa17071.pdf>
<https://www.nhtsa.gov/file-downloads?p=nhtsa%2Fdownloads%2FCRSS%2F>
<https://afdc.energy.gov/data/10380>
<https://ar.casact.org/commercial-auto-woes-what-will-it-take-to-make-the-line-profitable/>

Input Mileage Estimation

Within our assumed inputs, we do not directly obtain the mileage. As a result, we try to estimate annual mileage as a function of drivers, power units, and operating radius. Under the assumption that there are 250 operating days in a year, each round-trip on average is 1.1x the operating radius, and drivers are always operating power units when available, we estimate the annual mileage as $250 * \min(drivers, power\ units) * 1.1 * operating\ radius$.

Frequency Model

Machine Learning Models

We aim to estimate crash frequency per mile, which allows us to determine an annualized crash frequency from our quoting inputs.

Truck crash frequency data was taken from FMCSA's "Crash File". We fit both Generalized Linear Models (GLMs) and Gradient Boosted Trees on input data available at the time of quote (# power units, carrier operation type, cargo type).

From the input provided, both models showed equal predictive power, and thus the GLM was chosen due to simplicity and interpretability. This also has advantages should regulatory reporting be required for this pricing line.

The Poisson GLM model showed a 98.9% Actual/Estimated in crash counts, along with the predicted rates being within 0.6% of the actual for all groups of predictors.

Credibility Factor

Another consideration for frequency is the existence of prior crashes. To account for prior data, we take a weighted average of the carrier's historical crash frequency (if available) and the baseline frequency calculated from the model. As a result of the Poisson distribution's standard error convergence, we take the weight of the historical frequency to be

$$\sqrt{HISTORICAL\ MILEAGE / MILEAGE\ for\ FULL\ CREDIBILITY}.$$

Mileage for full credibility was determined as 1,000,000. Since we assume an individual power unit cannot travel over 100,000 miles in a year, the most possible mileage a carrier can have is 500,000. Under the formula above, this would put ~70% weight on the carrier's prior experience from public data, which appears reasonable.

Severity Model

Due to the lack of “insurance dollar” claim data, the Property Damage (PD) and Bodily Injury (BI) severities of primary auto liability coverage were estimated separately.

Property Damage

Due to data scarcity, fitting a proper model to PD severity was very difficult. As a result, we apply a factor to a “baseline” severity taken from TAIPA. The factor is just a linear function of the weight of the power units. The weight factor scales from 0.7 (when the power unit is under 10,000lbs) to 1.5 (when the power unit is at the maximum weight of 80,000lbs). This factor is then applied to a base severity of \$9,500 which is the trended TAIPA value at an annual rate of 5%, trended to 2027. The factors were taken purely out of judgment.

Bodily Injury

Similarly, fitting a model on BI severity proved difficult. The result was to use approximations from FHWA. In their report, they estimated the socio-economic costs of various levels of crash severities on the KABCO scale. These cost relativities were used as proxies for insurance claim severities as well.

To apply the KABCO scale, crash data was taken from the NHTSA, where we looked at the empirical distribution of crash severities separated by region (NorthEast, MidWest, South, West). This was used as a proxy for the loss distribution given a crash occurred.

Final Pricing

Premiums

After the frequency and severity models are calibrated, premium estimation is done once expense and profit provisions are decided. From a report by the CAS, we choose an expense rate to be ~30% and a profit margin of ~10%. It's noted that commercial auto lines have had underwriting losses for 14 years straight. As a result, if this is a new line for the company, it may make sense to be safer and set higher expense/profit margins, to lower the risk of severe losses at the cost of market competitiveness.

Declining Coverage

Due to lack of existing claims data, it was very difficult to find specific rules to decline. As such, I operated on the idea of declining quotes that either had bad historical evidence, or pose a threat to the book quality in the future from either input characteristics or calculated premium. Thus the following rules were decided:

- Annual mileage > 50,000 AND carrying hazardous materials
 - This seems very high risk to large liability losses
- Calculated premium is >3x the baseline premium for the inputted mileage

- This indicates the inputs are high risk given our pricing model
- Calibrated frequency (with historical data) is greater than 2x the baseline frequency
 - This indicates the carrier has bad historical experience, likely leading to being a high risk account

Additional Considerations

Given the BI severities are very similar, the book would likely attract smaller trucks from carriers with few historical crashes or no data. Since we are pricing a primary auto liability book, this could be profitable since smaller trucks pose less of a threat to other vehicles and pedestrians, thus resulting in a potentially lower claim severity.

The major issue with the model is definitely the severity estimations. As we obtain more claims data, we can refine the severity model as needed.

Future changes include input parameter modifications as well as better datasets to allow for better modelling.